Experiment 2 - Upload and analyze the data set given in CSV format, and perform data preprocessing and visualization.

Aim:

To upload and analyze the data set given in CSV format and perform data pre-processing and visualization.

Program:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from google.colab import drive
drive.mount('/content/drive')
file_path='/content/drive/MyDrive/sales_data (1).csv'
df=pd.read_csv (file_path)

# Display the first few rows of the DataFrame
print(df.head())

# Check for missing values
print(df.isnull().sum())

# Fill or drop missing values if necessary
df['Sales'] = df['Sales'].fillna(df['Sales'].mean())
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)

# Summary statistics
print(df.describe())
```

```
# Group by product and calculate the total sales and quantity
product_summary = df.groupby('Product').agg({
'Sales': 'sum',
'Quantity': 'sum'
}).reset_index()
print(product_summary)

# Bar plot of total sales by product
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()

# Line plot of sales over time
df['Date'] = pd.to_datetime(df['Date'])
sales_over_time = df.groupby('Date').agg({'Sales': 'sum'}).reset_index()
plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'],sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('SalesOver Time')
plt.show()
pivot_table = df.pivot_table(values='Sales', index='Region', columns='Product',
aggfunc='sum', fill_value=0)
print(pivot_table)
correlation_matrix = df.corr(numeric_only=True)
print(correlation_matrix)
import seaborn as sns
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
```

```
plt.title('Correlation Matrix')
plt.show()
```

Output:

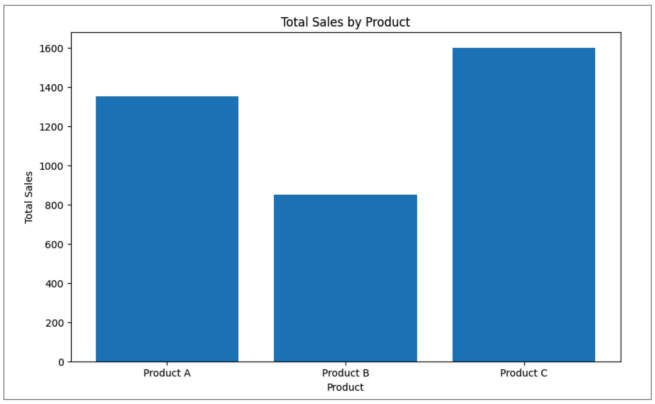
	Date	Product	Sales	Quantity	Region
0	2023-01-01 00:00:00	Product A	200	4	North
1	2023-02-01 00:00:00	Product B	150	3	South
2	2023-03-01 00:00:00	Product A	220	5	North
3	2023-04-01 00:00:00	Product C	300	6	East
4	2023-05-01 00:00:00	Product B	180	4	West
5	2023-06-01 00:00:00	Product A	210	5	North
6	2023-07-01 00:00:00	Product C	320	7	East
7	2023-08-01 00:00:00	Product B	160	3	South
8	2023-09-01 00:00:00	Product A	230	6	North
9	2023-10-01 00:00:00	Product C	310	7	East
10	2023-11-01 00:00:00	Product B	190	4	West
11	2023-12-01 00:00:00	Product A	240	6	North
12	13-01-2023	Product C	330	8	East
13	14-01-2023	Product B	170	3	South
14	15-01-2023	Product A	250	7	North
15	16-01-2023	Product C	340	8	East

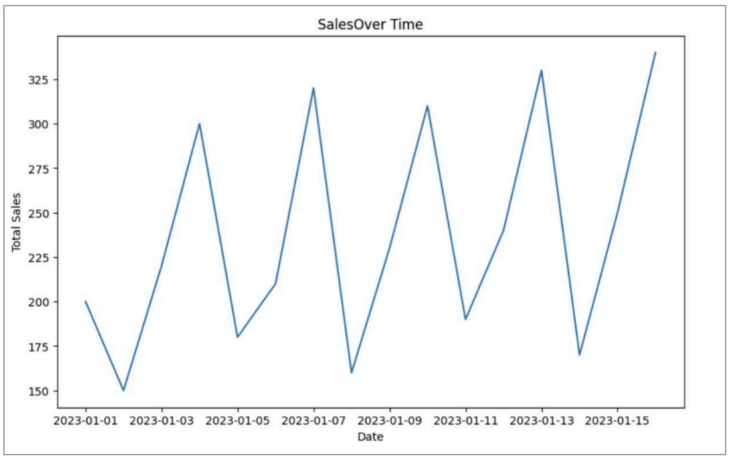
	Date	Product	Sales	Quantity	Region
0	2023-01-01 00:00:00	Product A	200	4	North
1	2023-02-01 00:00:00	Product B	150	3	South
2	2023-03-01 00:00:00	Product A	220	5	North
3	2023-04-01 00:00:00	Product C	300	6	East
4	2023-05-01 00:00:00	Product B	180	4	West

```
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64
```

	Sales	Quantity
count	16.000000	16.000000
mean	237.500000	5.375000
std	64.031242	1.746425
min	150.000000	3.000000
25%	187.500000	4.000000
50%	225.000000	5.500000
75%	302.500000	7.000000
max	340.000000	8.000000

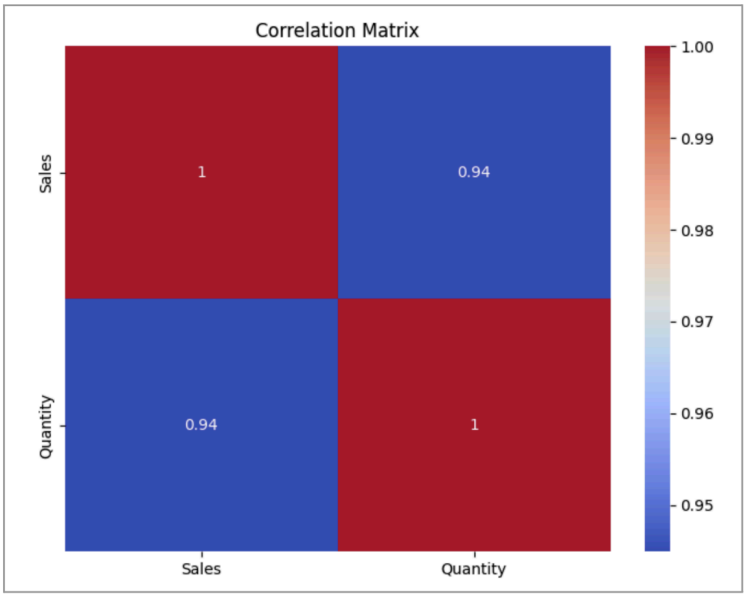
	Product	Sales	Quantity
0	Product A	1350	33
1	Product B	850	17
2	Product C	1600	36





Product	Product A	Product B	Product C
Region			
East	0	0	1600
North	1350	0	0
South	0	480	0
West	0	370	0

	Sales	Quantity
Sales	1.000000	0.944922
Quantity	0.944922	1.000000



Result:

Thus, we have uploaded and analyzed the CSV dataset and performed data preprocessing and visualization.